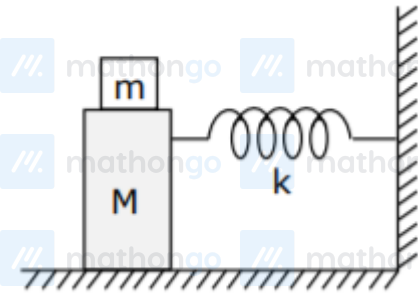


Questions with Answer Keys

MathonGo

Q1: 24 Feb (Shift 1) - Single Correct

In the given figure, a mass M is attached to a horizontal spring which is fixed on one side to a rigid support. The spring constant of the spring is k . The mass oscillates on a frictionless surface with time period T and amplitude A . When the mass is in equilibrium position, as shown in the figure, another mass m is gently fixed upon it. The new amplitude of oscillation will be -



- (1) $A\sqrt{\frac{M}{M+m}}$
- (2) $A\sqrt{\frac{M}{M-m}}$
- (3) $A\sqrt{\frac{M-m}{M}}$
- (4) $A\sqrt{\frac{M+m}{M}}$

Q2: 24 Feb (Shift 2) - Single Correct

When a particle executes SHM, the nature of graphical representation of velocity as a function of displacement is :

- (1) elliptical
- (2) parabolic
- (3) straight line
- (4) circular

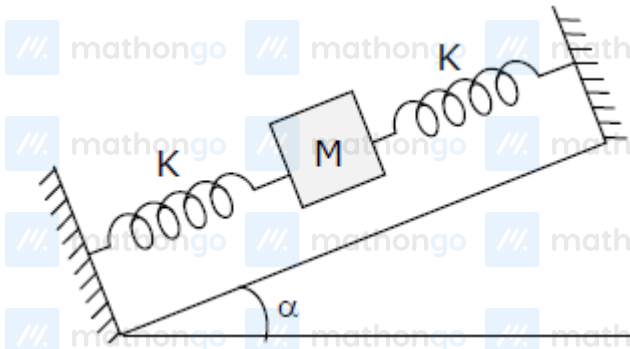
Q3: 24 Feb (Shift 2) - Single Correct

In the given figure, a body of mass M is held between two massless springs, on a smooth inclined plane. The free ends of the springs are attached to firm supports. If each spring has spring constant k , the frequency of

Questions with Answer Keys

MathonGo

oscillation of given body is :



- (1) $\frac{1}{2\pi} \sqrt{\frac{2k}{Mg \sin \alpha}}$
- (2) $\frac{1}{2\pi} \sqrt{\frac{k}{Mg \sin \alpha}}$
- (3) $\frac{1}{2\pi} \sqrt{\frac{2k}{M}}$
- (4) $\frac{1}{2\pi} \sqrt{\frac{k}{2M}}$

Q4: 24 Feb (Shift 2) - Single Correct

A particle is projected with velocity v_0 along x-axis. A damping force is acting on the particle which is proportional to the square of the distance from the origin i.e. $ma = -\alpha X^2$. The distance at which the particle stops :

- (1) $\left(\frac{2v_0}{3\alpha}\right)^{\frac{1}{3}}$
- (2) $\left(\frac{3v_0^2}{2\alpha}\right)^{\frac{1}{2}}$
- (3) $\left(\frac{3v_0^2}{2\alpha}\right)^{\frac{1}{3}}$
- (4) $\left(\frac{2v_0^2}{3\alpha}\right)^{\frac{1}{2}}$

Q5: 25 Feb (Shift 1) - Single Correct

If the time period of a two meter long simple pendulum is 2 s, the acceleration due to gravity at the place where pendulum is executing S.H.M. is :

- (1) $2\pi^2 \text{ ms}^{-2}$

Questions with Answer Keys

MathonGo

(2) 16 m/s^2

(3) 9.8 ms^{-2}

(4) $\pi^2 \text{ ms}^{-2}$

Q6: 25 Feb (Shift 2) - Single Correct

$Y = A \sin(\omega t + \phi_0)$ is the time - displacement equation of a SHM, At $t = 0$ the displacement of the particle is $Y = \frac{A}{2}$ and it is moving along negative x -direction. Then the initial phase angle ϕ_0 will be

(1) $\frac{\pi}{6}$

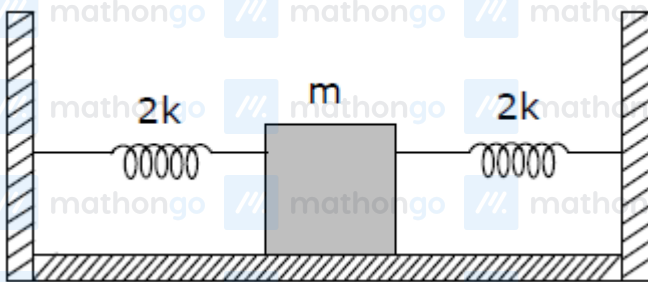
(2) $\frac{\pi}{3}$

(3) $\frac{2\pi}{3}$

(4) $\frac{5\pi}{6}$

Q7: 25 Feb (Shift 2) - Single Correct

Two identical spring of spring constant ' $2K$ ' are attached to a block of mass m and to fixed support (see figure). When the mass is displaced from equilibrium position on either side, it executes simple harmonic motion. Then time period of oscillations of this system is:



(1) $\pi \sqrt{\frac{m}{k}}$

(2) $\pi \sqrt{\frac{m}{2k}}$

(3) $2\pi \sqrt{\frac{m}{k}}$

(4) $2\pi \sqrt{\frac{m}{2k}}$

Q8: 26 Feb (Shift 1) - Single Correct

Questions with Answer Keys

MathonGo

Assume that a tunnel is dug along a chord of the earth, at a perpendicular distance $(R/2)$ from the earth's centre, where 'R' is the radius of the Earth. The wall of the tunnel is frictionless. If a particle is released in this tunnel, it will execute a simple harmonic motion with a time period :

(1) $2\pi\sqrt{\frac{R}{g}}$

(2) $\frac{1}{2\pi}\sqrt{\frac{g}{R}}$

(3) $\frac{2\pi R}{g}$

(4) $\frac{g}{2\pi R}$

Q9: 26 Feb (Shift 1) - Single Correct

If two similar springs each of spring constant K_1 are joined in series, the new spring constant and time period would be changed by a factor :

(1) $\frac{1}{2}, \sqrt{2}$

(2) $\frac{1}{4}, 2\sqrt{2}$

(3) $\frac{1}{2}, 2\sqrt{2}$

(4) $\frac{1}{4}, \sqrt{2}$

Q10: 26 Feb (Shift 2) - Single Correct

A particle executes S.H.M., the graph of velocity as a function of displacement is:

(1) a circle

(2) a parabola

(3) an ellipse

(4) a helix

Q11: 26 Feb (Shift 2) - Single Correct

Given below are two statements :

Statement (I) :- A second's pendulum has a time period of 1 second.

Questions with Answer Keys

MathonGo

Statement (II) :- It takes precisely one second to move between the two extreme positions.

In the light of the above statements, choose the correct answer from the options give below.

- (1) Both Statement I and Statement II are false
- (2) Statement I is true but Statement II is false
- (3) Statement I is false but Statement II is true
- (4) Both Statement I and Statement II is true

Q12: 26 Feb (Shift 2) - Numerical

A particle excutes S.H.M with amplitude 'a' and time period T . The displacement of the particle when its speed is half of maximum speed is $\frac{\sqrt{x}a}{2}$. The value of x is

Q13: 26 Feb (Shift 2) - Numerical

Time period of a simple pendulum is T . The time taken to complete $\frac{5}{8}$ oscillations starting from mean position is $\frac{\alpha}{\beta} T$. The value of α is

Questions with Answer Keys

MathonGo

Answer Key

Q1 (1)

Q2 (1)

Q3 (1)

Q4 (3)

Q5 (1)

Q6 (4)

Q7 (1)

Q8 (1)

Q9 (1)

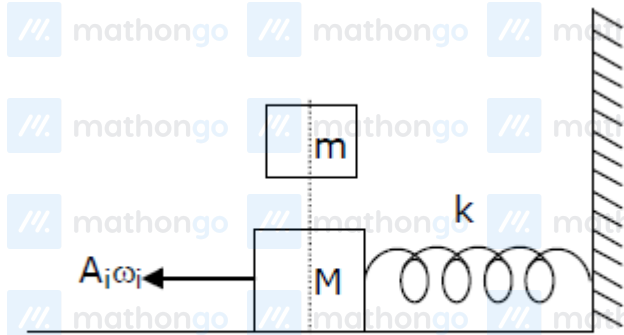
Q10 (3)

Q11 (3)

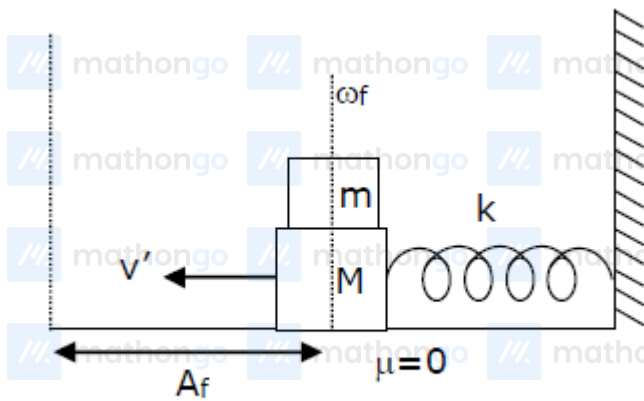
Q12 (3)

Q13 (7)

Q1 (I)



Before placing



After placing

We know that $\omega = \sqrt{\frac{k}{m}}$ and $\omega_i = \sqrt{\frac{k}{M}}$ $A_i = A$

Also, momentum is conserved just before and just after the block of mass (m) is placed because there is no impulsive force. So -

$$MA_i\omega_i = (M+m)v'$$

$$v' = \frac{MA_i\omega_i}{(M+m)} \Rightarrow v' = A_f\omega_f$$

$$\frac{MA\omega_i}{(M+m)} = A_f\sqrt{\frac{K}{(M+m)}}$$

$$\Rightarrow \frac{MA\sqrt{\frac{K}{M}}}{M+m} \times \sqrt{\frac{M+m}{K}} = A_f$$

$$\Rightarrow A_f = A\sqrt{\frac{M}{(M+m)}}$$

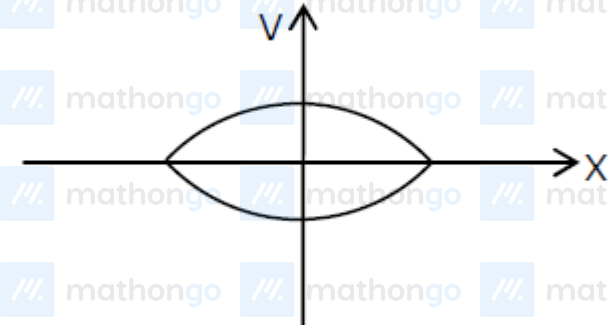
Hints and Solutions

MathonGo

Q2 (1)

We know that is SHM:

$$V = \omega \sqrt{A^2 - x^2}$$



elliptical

Q3 (1)

Equivalent $K = K + K = 2K$

$$\text{Now, } T = 2\pi \sqrt{\frac{m}{K_{eq}}}$$

$$\Rightarrow T = 2\pi \sqrt{\frac{m}{2k}}$$

$$\therefore f = \frac{1}{2\pi} \sqrt{\frac{2k}{m}}$$

Q4 (3)

$$a = \frac{v dv}{dx}$$

$$\int_{v_i}^{v_f} V dv = \int_{x_i}^{x_f} a dx$$

Given : $-v_i = v_0$

$$V_f = 0$$

$$X_i = 0$$

$$X_f = x$$

From Damping Force : $a = -\frac{\alpha x^2}{m}$

$$\int_{V_0}^0 V dV = - \int_0^x \frac{\alpha x^2}{m} dx$$

Hints and Solutions

MathonGo

$$-\frac{v_0^2}{2} = \frac{-\alpha}{m} \left[\frac{x^3}{3} \right]$$

$$x = \left[\frac{3mv_0^2}{2\alpha} \right]^{\frac{1}{3}}$$

Q5 (1)

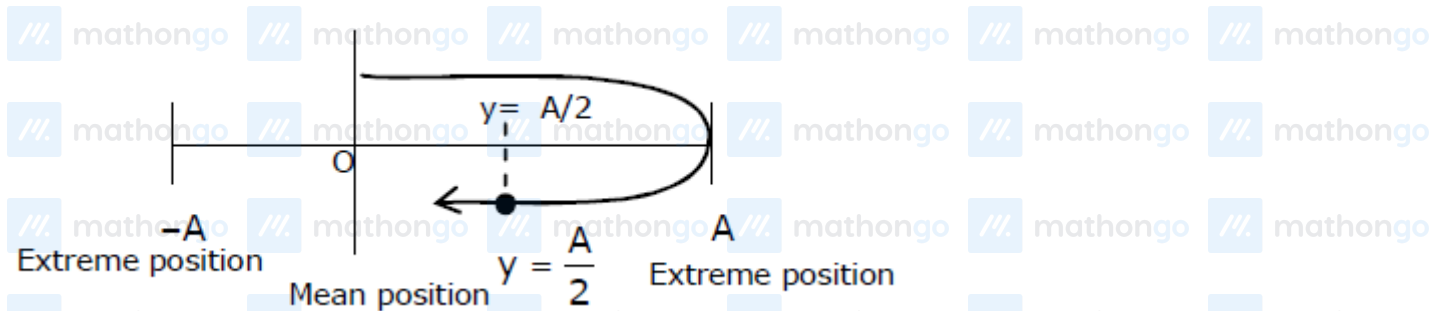
$$T = 2\pi \sqrt{\frac{I}{g}}$$

$$T^2 = \frac{4\pi^2 I}{g}$$

$$g = \frac{4\pi^2 I}{T^2}$$

$$= \frac{4\pi^2 \times 2}{(2)^2} = 2\pi^2 \text{ ms}^{-2}$$

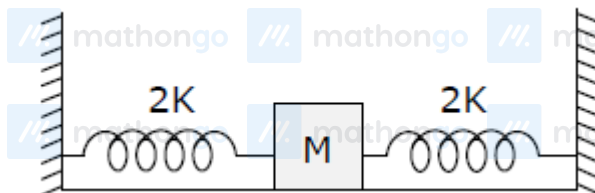
Q6 (4)



The initial phase angle $\phi_0 = \pi - \frac{\pi}{6}$

$$= \frac{5\pi}{6}$$

Q7 (1)



Hints and Solutions

MathonGo

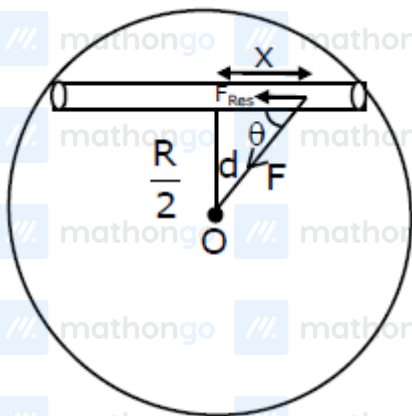
Due to parallel combination $K_{\text{eff}} = 2k + 2k$

$$\therefore T = 2\pi \sqrt{\frac{m}{k_{\text{eff}}}}$$

$$= 2\pi \sqrt{\frac{m}{4k}}$$

$$T = \pi \sqrt{\frac{m}{k}}$$

Q8 (1)



$$\cos \theta = \frac{x}{d}$$

If displaced from equilibrium position,

$$F_{\text{restoring}} = \left(\frac{GMmd}{R^3} \right) \cos \theta$$

$$F_{\text{Res.}} = \frac{GMmd}{R^3} \cdot \frac{x}{d} = \frac{GMmx}{R^3}$$

$$a_R = \frac{GMx}{R^3} \quad GM_e = gR^2$$

$$T = 2\pi \sqrt{\left| \frac{x}{a} \right|}$$

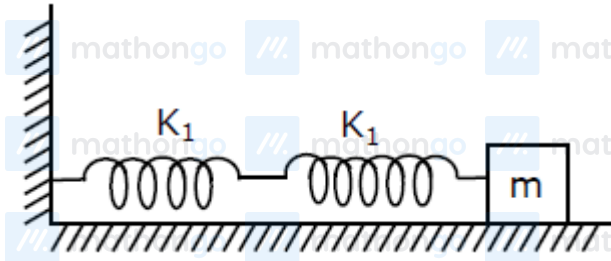
$$T = 2\pi \sqrt{\frac{x}{GMx}} \sqrt{\frac{R^3}{gR^2}}$$

$$T = 2\pi \sqrt{\frac{R}{g}}$$

Q9 (1)



$$T = 2\pi \sqrt{\frac{m}{K_1}}$$



Q10 (3)

For a body performing SHM, relation between velocity and displacement $v = \omega \sqrt{A^2 - x^2}$

now, square both side $v^2 = \omega^2 (A^2 - x^2)$

$$\Rightarrow v^2 = \omega^2 A^2 - \omega^2 x^2$$

$$v^2 + \omega^2 x^2 = \omega^2 A^2$$

divide whole equation by $\omega^2 A^2$ $\frac{v^2}{\omega^2 A^2} + \frac{\omega^2 x^2}{\omega^2 A^2} = \frac{\omega^2 A^2}{\omega^2 A^2}$

$$\frac{v^2}{(\omega A)^2} + \frac{x^2}{A^2} = 1$$

above equation is similar as standard equation of ellipses, so graph between velocity and displacement will be ellipses.

Q11 (3)

As we know time period of second's pendulum is 2sec, so statement (1) is incorrect.

Time taken between two extreme points in second's pendulum is 1 sec.

Above statement is correct because time taken by particle performing SHM between two extreme position is $T/2$.

Here, $T = 2$ sec.

So, time $= 2/2 = 1$ sec

Q12 (3)

For a particle executes S.H.M

$$V = \omega \sqrt{a^2 - x^2}$$

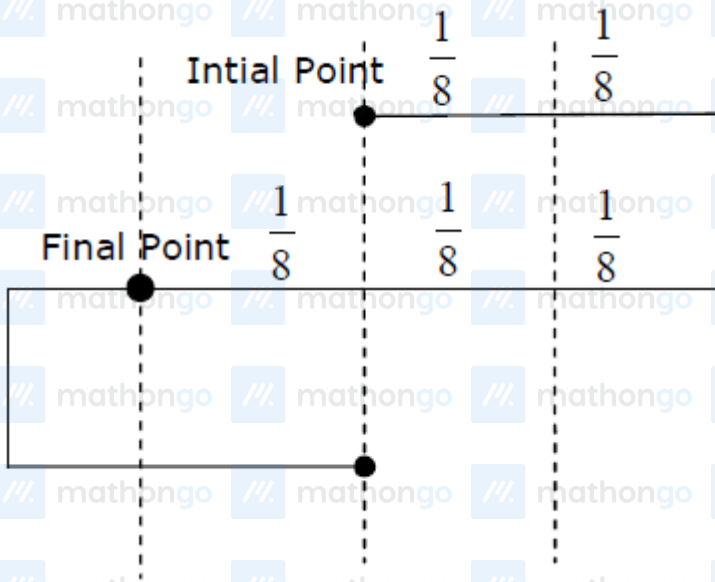
$$\text{Given } V = \frac{V_{\max}}{2} \Rightarrow \frac{A\omega}{2}$$

$$\frac{A^2\omega^2}{4} = \omega^2 a^2 - \omega^2 x^2$$

$$x = \frac{\sqrt{3}}{2} a$$

Q13 (7)

For given $\left(\frac{5}{8}\right)$ oscillation, we can write it as $\rightarrow \left(\frac{1}{2} + \frac{1}{8}\right)$ And we know for half oscillations time $\rightarrow \frac{T}{2}$



$$\text{For final point } \rightarrow \pi + \frac{\pi}{6} \Rightarrow \frac{7\pi}{6}$$

$$\text{Time } \rightarrow \frac{7T}{12} \rightarrow \text{given } \rightarrow \frac{\alpha}{\beta} T \alpha = 7p$$